

Deliang Wei

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 Johns Hopkins University  ECE & DSAI • Postdoctoral Researcher  1997-05-10

 Baltimore, Maryland, United States

I received my Ph.D. in Mathematics, with a research direction in image processing, from the School of Mathematical Sciences at East China Normal University in July 2025, under the supervision of Professor Fang Li. I am currently a postdoctoral researcher at the DSAI Institute, Johns Hopkins University, working with Professor Yu Sun. My research focuses on theoretically grounded generative AI methods for computational imaging. I have published multiple first-author papers in top-tier international machine learning conferences and applied mathematics journals. My research background covers generative AI models, optimization algorithms, monotone operator theory, posterior sampling algorithms, and related mathematical tools.

Research Interests

- › Applied mathematics: operator theory, spectral theory, optimization, posterior sampling theory, and convergence analysis of algorithms.
- › Deep learning: diffusion-type methods with convergence guarantees, posterior sampling, and plug-and-play algorithms.
- › Parallelization and acceleration: acceleration of iterative algorithms based on deep learning models and multi-GPU parallel algorithms.
- › Applications: computational imaging, including imaging inverse problems, medical imaging, and radar scattering imaging.

Education and Research Experience

August 2028	Whiting School of Engineering, Johns Hopkins University
August 2025	Postdoctoral Researcher, Advisor: Professor Yu Sun in ECE & DSAI
July 2025	School of Mathematical Sciences, East China Normal University
September 2019	Ph.D., Advisor: Professor Fang Li in Computational Mathematics
February 2023	Department of Mathematics, Chinese University of Hong Kong
December 2022	Visiting Scholar, Advisor: Professor Tieyong Zeng in Research Assistant
July 2019	School of Mathematical Sciences, East China Normal University
September 2015	B.Sc. in Mathematics and Applied Mathematics

Publications

[* denotes first author]

- › Renaud M*, Hermant J*, **Wei D***, and Sun Y. Provably Accelerated Imaging with Restarted Inertia and Score-based Image Priors[C]. *The Fourteenth International Conference on Learning Representations*, **ICLR** 2026.
- › **Wei D***, Chen P, Xu H, Yao J, Li F, and Zeng T. Learning Cocoercive Conservative Denoisers via Helmholtz Decomposition for Poisson Imaging Inverse Problems[C]. *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, **NeurIPS** 2025.
- › **Wei D***, Chen P, and Li F. Learning pseudo-contractive denoisers for inverse problems[C]. *Proceedings of the 41st International Conference on Machine Learning*, PMLR, 235:52500–52524, **ICML** 2024.
- › **Wei D***, Li F, Shen X, and Zeng T. DeepSPIM: Deep Semi-Proximal Iterative Method for Sparse-View CT Reconstruction with Convergence Guarantee[J]. *CSIAM Transactions on Applied Mathematics*, 5(3): 421–447,

CSIAM-AM 2024.

- › Chen P*, Li F, **Wei D**, and Lv C. Spatiotemporal traffic data completion with truncated minimax-concave penalty[J]. *Transportation Research Part C: Emerging Technologies*, 164: 104657, **TRC** 2024.
- › Chen P*, Li F, **Wei D**, and Lv C. Low-Rank and Deep Plug-and-Play Priors for Missing Traffic Data Imputation[J]. *IEEE Transactions on Intelligent Transportation Systems*, **IEEE TITS** 2024.
- › **Wei D***, Weng S*, and Li F. Nonconvex Rician noise removal via convergent plug-and-play framework[J]. *Applied Mathematical Modelling*, 123: 197–212, **AMM** 2023.
- › **Wei D***, Li F, and Weng S. Cauchy Noise Removal via Convergent Plug-and-Play Framework with Outliers Detection[J]. *Journal of Scientific Computing*, 96(3): 76, **JSC** 2023.
- › **Wei D***, Chen P, Li F, and Zhang X. Efficient second-order optimization with predictions in differential games[J]. *Optimization Methods and Software*, 1–26, **OMS** 2023.
- › **Wei D*** and Li F. Flexible parameter selection methods for Rician noise removal with convergence guarantee[J]. *International Journal of Computer Mathematics*, 99(11): 2250–2271, **IJCM** 2022.
- › **Wei D***, Li F, and Zeng T. A Low-Dose Computed Tomography Imaging Method Based on a Semi-Proximal Operator. Invention patent, under review.

Preprints

[* denotes first author]

- › **Wei D***, Bell E, Chen Y, and Sun Y. Towards Scalable Generative Imaging: Parallel Proximal Sampling via Picard Iterations. Submitted to *JMLR*.
- › Yao J*, **Wei D**, and Li F. Envelope Forms for Nonconvex ℓ_0 Surrogates in Sparse Optimization. Submitted to *Inverse Problems*.
- › **Wei D***, Chen P, and Li F. Convergent Ishikawa Methods with Deep Pseudo-Contractive Denoisers. In preparation.
- › Chen P*, **Wei D***, Yao J, and Li F. Tensor Correlated Total Variation Meets Deep Pseudo-Contractive Denoisers: Tensor Completion via Monotone Inclusion. Under review at *IEEE TNNLS*.

Honors and Research Projects

- › National scholarship for doctoral students, China, 2024.
- › East China Normal University Academic Innovation Promotion Program for Excellent Doctoral Students, YBNLTS2024-018: “A spectral regularization method for neural networks via holomorphic functional calculus”.
- › Selected for the 2024 China Association for Science and Technology (CAST) Young Talent Support Program - Special Project for Doctoral Students, supported by China Computer Federation (CCF) society.

Professional Activities

- › Reviewer for top-tier machine learning and computational imaging conferences and journals, including NeurIPS, ICLR, ICML, AAAI, ECCV, CVPR, IEEE TCI, and TMLR.
- › Invited seminar talk at the Computational Imaging Research Group, Johns Hopkins University, December 12, 2024.
Title: “Plug-and-Play Ishikawa Process with Pseudo-Contractive Denoisers for Imaging Inverse Problems.”

Teaching Experience

Present October 2025	Seminar Organizer @ Whiting School of Engineering, Johns Hopkins University Organized a seminar on probability and sampling theory, with regular discussions on theoretical topics related to learning and sampling.
May 2026 January 2026	Teaching Assistant @ Whiting School of Engineering, Johns Hopkins University Teaching assistant for Professor Yu Sun's optimization course; responsible for homework grading, exam grading, and office hours.
December 2023 September 2023	Seminar Organizer @ School of Mathematical Sciences, East China Normal University Organized a seminar on monotone operator theory, with regular discussions on monotone operator theory and related analysis.
June 2021 September 2020	Teaching Assistant and Problem Session Instructor @ School of Mathematical Sciences, East China Normal University - Teaching assistant for the honors Mathematical Analysis course taught by Professors Tao Cheng and Xuecheng Pang. - Responsible for homework grading, exam grading, problem-session support, and student Q&A.

Skills

- Languages: Chinese (native); English (proficient).
- Tools: PyTorch, Linux, MATLAB, Git, SSH.

References

- Professor Yu Sun, Postdoctoral Advisor.
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